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Digital twin-assisted flexible slice admission control for 5G core network: A deep reinforcement learning approach

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ABSTRACT

5G core network slicing is an important component of 5G end-to-end slicing, which divides the physical network into multiple logical networks to handle various critical service flows, and brings better cost effectiveness to operators. Due to the variety of services and the limited resources of the core network, slice admission control is particularly important, which can selectively accept or reject the slice request, so as to maximize the operators' revenue. However, existing works mainly focus on the complete acceptance or rejection of the slice request, and rarely consider the partial admission. In fact, there exists the best-effort slice in the network, which requires relaxed service performance and its resources can be appropriately adjusted. In view of this, we consider partial admission of slice requests under limited resources, introduce slice scaling factor to characterize the resource adjustment threshold, and propose a flexible slice admission control (FSAC) scheme for 5G core network based on deep reinforcement learning (DRL) with the goal of improving slice deployment revenue. In order to avoid the negative impact on real physical network caused by DRL agent's policy exploration, we also design a digital twin-assisted slice admission control framework. Finally, experimental results demonstrate the effectiveness of the proposed scheme in improving slice deployment revenue and acceptance ratio.

1. Introduction

With the continuous advancement of 5G construction, network slicing technology, which can provide diverse services for various application scenarios, has become a research hotspot in academia and industry. 5G core network slicing is the key component between the access and the data networks, which customizes network functions through virtualization for multi-tenants on the service-based architecture, and handles various critical service flows including control and management [1,2]. When a core network slice is activated, the resources in the core network infrastructure will be allocated to each network function that makes up the slice, and the resource occupancy will continue until the slice is decommissioned [3]. Therefore, although core network slicing has the potential to meet diversified service requirements and realize customized service enhancement, due to the existence of numerous service flows, it is difficult for the network operator to fully accept all slice establishment requests (slice requests for short) with limited resources. Instead, the operator needs to selectively accept or reject new slice requests based on the available resources to guarantee its revenue [4,5]. At this time, effective slice admission control is particularly important.

Many efforts have been made directly to slice admission control. For example, the authors in [6] propose an efficient 5G core network slice admission control and resource allocation method, which considers the quality of service (QoS) requirements of typical 5G use cases to process slice requests and improve the revenue of service providers. A slicing prioritized admission control mechanism based on infrastructure resource reservation is investigated in [7], which can deal with the dynamic nature of slice request and the uncertainty of resource demand. A reinforcement learning (RL)-based slice deployment and orchestration framework is presented in [8] to maximize resource utilization and operators' revenue, with an online priority-based algorithm that controls the slice admission. These works, although provide excellent solutions, mainly focus on the complete acceptance or rejection of the slice request, and rarely consider partial slice admission. In the case of complete slice admission, some slice requests may be rejected for lack of a small amount of resource. However, scaling down the requested resources and accepting them may have little impact on the service performance. In fact, there are best-effort slices in the network, which require relaxed service performance and do not rely

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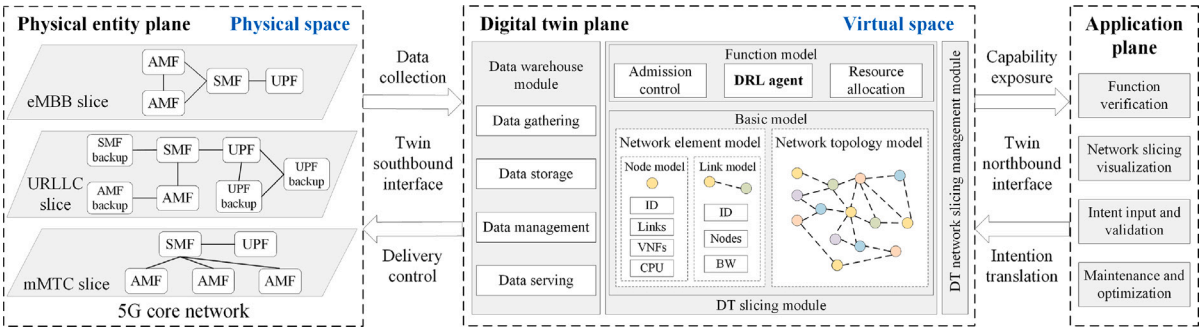


Fig. 1. Digital twin-assisted slice admission control framework for 5G core network.

on dedicated resources. Its requirement can be appropriately adjusted according to the network conditions [8,9]. Flexible admission control of this type of slice, that is, adjusting and determining the proportion of requested resources of the slice request, is the key to improving the slice acceptance ratio and operators' revenue, and is also the focus of this paper.

The partial admission of slice request actually reflects the elasticity of slicing and the flexibility of resource allocation. Many standards developing organizations such as 3GPP (the 3rd Generation Partnership Project) and ETSI (European Telecommunications Standards Institute), have considered the significance of this topic. For example, the 3GPP technical specification TS 22.261 [10] specifies that 5G system shall allow operators to define and update the set of services and capabilities supported by network slicing, and states that flexible network operations are the mainstay of the 5G system. The ETSI group specification GS ENI 001 [11] introduces the concept of elastic slice, and defines it as a set of virtual network functions (VNFs) and related resources to support mobile services with non-stringent demands, allowing for graceful performance degradation, which is also in line with our slice request partial admission concept. In addition, there exist some orchestration platforms that can be used for slice request partial admission, such as 5G-MoNArch [12] for elastic slice resource management, and the ONAP (Open Network Automation Platform) for flexible slice orchestration [13].

Moreover, there are lots of works that can provide reference for slice admission control. In terms of architecture, a distributed and scalable 5G slicing-enabled software-defined networking-based core network architecture is proposed in [14] with the goal of improving the QoS of autonomous driving applications, and a network slicing model is also constructed to meet low latency constraints. For the resource allocation and management in 5G, in terms of transmission resource allocation, the authors in [15] minimize the overall energy consumption of 5G ultra-dense cellular networks by optimizing carrier allocation and power utilization. A novel approach to hierarchical resource allocation in uplink–downlink decoupled heterogeneous network is proposed in [16] to meet capacity and QoS demands. In terms of computational resource allocation, the authors in [17] propose a dynamic optimization model to maximize energy efficiency while satisfying the QoS constraints with consideration of the user fairness criteria. Although these works do not directly study slice admission control, they are of great significance for the design of architecture and resource allocation mechanisms in slice admission control.

Deep reinforcement learning (DRL) technology has been widely used in slice admission control and resource allocation [18,19]. Existing research shows that DRL has a strong ability to solve complex and high-dimensional network problems, and can dynamically adjust the decision through environment interaction and strategy update [20,21]. In the related work [6,8] mentioned above, DRL algorithms are involved for effective slice admission control. However, the DRL agent will produce lots of non-optimal decisions when it conducts trial-and-error exploration in environment interaction [22]. If the DRL model is directly deployed in the real physical core network as the brain and control center of the whole network for training, serious negative impact can be caused on the normal operation of the network, leading to network delays and even failures, which is generally unbearable for operators. Digital twin (DT) is an innovative technology to digitally depict physical entities and time-varying performance leveraging virtual replicas, and can be used to perform slice admission control scheme in various what-if scenarios without affecting the real physical core network [23,24]. By interacting with the physical core network in real time, DT for network slicing can greatly assist in DRL decision making and policy execution, paving the way for designing efficient and feasible slice admission control schemes [25]. Therefore, we apply DT in the slice admission control scheme to avoid the negative effects brought by DRL agent's trial-and-error exploration. The main contributions of this paper are summarized as follows.

- We consider partial admission of slice requests to make full use of network resources, introduce slice scaling factor to characterize the resource adjustment threshold, and propose a DRL-based flexible slice admission control (FSAC) scheme for 5G core network with the goal of improving slice deployment revenue.
- We design a DT-assisted 5G core network slice admission control framework, give the digital description of 5G core network slicing, and provide a virtual environment for DRL agent to interact, avoiding the damage to the normal operation of the physical network caused by DRL agent's policy exploration.
- Experimental results show that our proposed DRL-based slice admission control scheme has significant advantages in improving slice acceptance ratio and slice deployment revenue.

The rest of this paper is organized as follows. The DT-assisted slice admission control framework for 5G core network is provided in Section 2. Section 3 presents the proposed DRL-based slice admission control scheme. The performance of our scheme is evaluated in Section 4. Finally, Section 5 concludes this paper.

2. DT framework and performance metric

2.1. DT framework for slice admission control

Fig. 1 describes the DT-assisted slice admission control framework for the 5G core network, which consists of three parts including the physical entity plane, digital twin plane, and network application plane, with the reference to IETF (Internet Engineering Task Force) Internet-Draft documents [26,27]. For the physical entity plane, the network slicing involving multiple associated VNFs is deployed on the core network nodes to provide customized services for 5G use-cases, e.g., enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultrareliable low-latency communication (URLLC). These slices can have different requirements, including slice structures and resource requirements. For example, URLLC slices can be backed

up appropriately because they need to ensure high reliability, and mMTC slice needs to set up multiple access and mobility management functions (AMFs) to deal with massive device access.

The DT plane constructs a digital replica of the physical core network comprising data warehouse module, DT network slicing module (i.e., the combination of functional model and basic model), and DT slicing management module. The data warehouse module collects and stores the processing (represented by CPU in Fig. 1) and transmission (represented by BW in Fig. 1) resource condition of network nodes and connection links, and supports virtual network modeling based on data service [28]. In DT network slicing module, the basic model maps the physical nodes and links into virtual elements with a graph-structured topology representation, and updates the network status in real time to achieve accurate digital characterization. The function model leverages the intelligent DRL agent to explore the optimal slice admission control policy based on the monitored resource conditions. DT slicing management module is responsible for maintaining the entire life cycle of the twin entity. These three key modules interact with each other via the corresponding interface protocol to complete closed-loop simulation and optimization [27].

The network application plane inputs requirements to the DT network through the twin northbound interface, and deploys services at DT network via model instance, which enables various applications such as function verification, network visualization, slice intention input and validation, as well as maintenance and optimization with less impact on the physical core network.

2.2. Workflow of the DT framework

In a continuously operating network, it is not practical to shut down the system to perform DRL training or to take tentative responsive actions. We cannot allow DRL agent to play around with slice admission control decisions in the real core network, which greatly harms the user experience. At this point, it seems effective to restrict some illegal actions with a check module. A well-designed check module can indeed avoid illegal actions to a great extent. However, it is worth noting that a legal action does not mean that it is a good action, it may be a legal and relatively "bad" action that can bring performance degradation to the system. In the actual operation of the network, we expect that it will always maintain a good service state, without performance degradation. Therefore, the use of the check module alone is not sufficient. Also, excessive blocking of actions can lead to insufficient exploration, and some temporarily "bad" actions may be the path to some high reward, which makes it difficult to set a fixed check criterion to filter actions. This is where DT comes into play. It can provide a high-fidelity environment where DRL training cannot cause damage to the physical network, and achieve scene update and replay in an interactive virtual-real mapping and data-driven manner. Note that the DT environment is different from the traditional simulation platform, the former emphasizes the interaction between virtual and physical, can be updated in real time and achieve dynamic evolution, while the latter lacks the interaction interface, and the fidelity is relatively low.

When a slice request arrives, the application plane identifies and translates the tenant's intentions into optimization target with network constrains, and delivers it to the DT plane for network management. Specifically, the DT slicing management module calls the function model to obtain the optimization problem, and the function model turns to the admission control and resource allocation functions for initiating the DRL exploration. Then, the basic model updates the network element model and the virtual topology to create the twin network of the physical core network assisted by the data warehouse module. When the DRL agent fully explores in the virtual environment and obtains the optimal decision, the function model forwards the network resource configuration to the physical plane and adjusts its admission control policy. In this way, unacceptable actions and bad performance generated by DRL agent's exploration can be transferred to the DT environment, reducing the negative impact on the physical network.

2.3. Digital description of core network slicing

The DT models the 5G core network as an undirected graph $G = \{V, L\}$, where V is the set of physical network nodes and L is the set of physical connection links between network nodes. Each node $v_i \in V$ supplies the processing resource represented by p_i, and the link $l_{i,j} \in L$ has an available bandwidth $b_{i,j}$. The DT network topology graph has the same characteristics as the real physical core network. The DT models the type-n core network slice also as an undirected graph $G^n = \{V^n, L^n\}$, where V^n and L^n stand for the set of VNFs and virtual links that make up the slice. v_i^n represents the ith VNF in type-n slice, and $l_{i,j}^n \in L^n$ is the virtual link between v_i^n and v_j^n. The processing resource required by v_i^n is expressed as p_i^n, and the bandwidth resource required by the virtual link between v_i^n and v_j^n is characterized by $b_{i,j}^n$.

The network slice request is denoted by $sr = \{n, \rho, \beta\}$, where $n \in N$ indicates the associated slice type (e.g., 5G use-cases), which also corresponds to certain slice template, and N is the set of all slice types. Then, ρ is the processing resource required by each VNF of the slice, and β is the transmission resource required by each virtual link. The slice template contains the definition of the slice graph G^n, the revenue of deploying this type of slice, and the slice scaling factor λ_n ($0 < \lambda_n \leq 1$) characterizing the resource adjustment threshold. The slice scaling factor indicates the maximum degree to which the requested resources can be scaled down, and limits the resource adjustment range. To be more specific, let rs_n represent the amount of requested resources of a type-n slice, let ps_n represent the amount of resources available for the slice, if the slice is to be accepted, the constraint $\frac{ps_n}{rs_n} \geq \lambda_n$, must be satisfied. Here, the best-effort slice can have a λ of less than 1. This kind of slice has relaxed service performance requirements, does not need dedicated network resources, and its required resources can be flexibly adjusted. For example, eMBB and mMTC slices are generally considered to be resource reducible [8,9]. In contrast, URLLC slice has stringent resource requirement to ensure reliability and low latency, and its λ is fixed to 1. The exact value of λ is important to avoid the risk of unacceptable slice performance degradation and can be agreed upon by the operator and the tenant. In addition, we also define the slice request processing window, which represents the number of slice requests that can be processed simultaneously during slice admission control and has the size of W.

2.4. Performance metric

We define two metrics to evaluate our proposed slice admission control scheme for 5G core network, namely the slice acceptance ratio and the slice deployment revenue. These types of metrics are often used to evaluate the slice admission control schemes [6,7,9]. The former is the ratio of the number of successfully accepted slice requests to the slice request processing window size W, which is used to evaluate the performance of the scheme in accepting slices, and can be expressed as:

$$AR = \frac{\sum_{w=1}^{W}\sum_{n=1}^{N} c_w^n \cdot e_w^n}{W}, \qquad \forall c_w^n \in \{0,1\}, \forall e_w^n \in \{0,1\}, \tag{1}$$

where c_w^n represents whether a type-n slice request has arrived in the wth position of slice request processing window, and e_w^n indicates whether this request has been successfully accepted. In addition, the slice deployment revenue represents the profit of deploying slices in a processing window, which is used to evaluate the performance of the proposed scheme in terms of improving the operators' revenue, and can be expressed as:

$$RV = \sum_{w=1}^{W}\sum_{n=1}^{N}(rv^n + pr_w^n \cdot g^n) \cdot c_w^n \cdot e_w^n, \quad \forall c_w^n \in \{0,1\}, \forall e_w^n \in \{0,1\}, \tag{2}$$

where rv^n is the profit of accepting a type-n slice request, $pr_w^n \cdot g^n$ is the additional gain of deploying a particular slice. pr_w^n is the admission

proportion of the type-n slice in the ωth position of the processing window, and g^n is the gain factor that indicates how much the admission proportion of type-n slice is valued. This additional gain exists to distinguish between completely and partially accepted slice request, which vary in the service performance and the consumed network resources.

We also define a metric to evaluate the application effect of DT, namely the blocking ratio of unacceptable decisions. It means the fraction of unacceptable "bad" decisions produced by DRL exploration for a given number of episodes, which will be blocked by DT environment such that negative effects on the physical network are avoided. The blocking ratio of unacceptable decisions can be represented as:

$$BR = \frac{\sum_{\varepsilon=1}^{EP} dt_\varepsilon * \mu_\varepsilon}{EP}, \forall dt_\varepsilon \in \{0,1\}, \mu_\varepsilon \in \{0,1\}, \tag{3}$$

where EP is the total number of episodes used for evaluation. dt_ε indicates whether the decision made in the εth episode is executed in DT environment. μ_ε indicates whether the decision is unacceptable, which can be determined based on the gap between the decision to be judged and the optimal decision.

3. DRL-based flexible slice admission control scheme for 5G core network

3.1. Problem formulation

In 5G core network, various tenants send slice requests to establish slices and obtain customized services. Due to the limited resources of the physical network, how to select and accept various types of slices to guarantee the operators' revenue is the key problem. Most of the existing solutions consider the complete acceptance and rejection of the slice. Actually, there exist best-effort slices in the network that have relaxed performance requirements. When the network is faced with insufficient resources, the requested resources of this kind of slice can be adjusted. Once the requested resources of a few slices have been partially accepted and appropriately scaled down, one or more new slice requests can be accommodated. Therefore, the problem we consider is how to flexibly perform partial admission control on slice requests to improve operators' revenue, that is to maximize the slice deployment revenue RV (see Eq. (2)).

The problem we try to solve is a slice admission control problem, which is to make slice admission decisions for incoming slice requests over a slice request processing window, such as whether to accept the slice request, whether to scale down the requested resource and accept it, and whether to reject the request. Lots of literature such as [6,29] have proved that the slice admission control process conforms to Markov decision process (MDP), which can be represented by a quadruple $\langle S, A, T, R\rangle$, where S and A are the sets of states and actions, T refers to the transition probability from all states to all subsequent states, and R is the reward function.

The problem we consider is also applicable. S can be the set of network environment information, including remaining resources, slice request status, and requests acceptance status; A can be the set of adjustments to the slice admission decision, such as to increase or decrease the admission proportion; the transition T means that under certain network conditions and slice request conditions, after some actions, the network conditions and slice request conditions change and reach the next state; R can be defined directly as the slice deployment revenue RV, since it is our optimization goal. The step-by-step exploration of slice admission control can be viewed as a sequential decision problem. Once the environment state is determined, the subsequent admission decision will not be affected by the environment state and decision before, which conforms to the Markov property. Therefore, our problem is suitable for applying RL that can solve MDP. Because of the variable and dynamic network state, it is difficult to describe the state transition matrix explicitly, so it is appropriate to adopt the model-free RL method that does not depend on the state transition probability. In the following, we further elaborate on RL framework, model-free deep Q-learning (DQL)-based FSAC scheme, and detailed definitions of state, action, and reward.

3.2. RL framework

RL is an important branch of machine learning, which is used to describe and solve the problem of agents learning policies to reach a specific goal during the interactions with the environment [30]. These problems can often be formalized as MDP, in which explicit transition probability, state space, and reward function are not necessarily required [31]. Given the state s_t of an environment, the agent will select a corresponding action a_t according to a certain policy, and the environment will change after the execution of a_t, that is, the state will be transformed into a new state s_{t+1}. After executing each action, the agent will get a reward r_t and adjust its policy according to this incentive value, aiming to obtain the maximum long-term cumulative reward $R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$, where γ is the discount coefficient denoting the degrees of attention to future rewards.

Specifically, the RL agent can be deployed as part of the function model in the DT plane, as shown in Fig. 1. After making each slice admission decision, the slice deployment status and the amount of available resources in the DT plane change, and the agent recollects new environment information and makes new decisions. Here, the agent learns to choose the action that maximizes RV based on the state and reward obtained from the interactive virtual environment and avoid damage to the physical network.

3.3. DRL-based slice admission control scheme

In RL, value-based methods are suitable for solving problems with discrete action spaces, where each action is explicitly evaluated and an action is selected at each step based on these evaluations. The proposed slice admission control scheme intends to adopt discrete actions, including different adjustment strategies to the slice admission condition, so it is appropriate for implementation by value-based RL methods. Note that while slice adjustment can be designed as a continuous action space and solved using a policy-based algorithm, discrete action space makes it easier to convert actions into resource adjustment commands. For example, if a bit of the action vector is positive one (see the detailed action definition in the following), it can be converted to allocating one more virtual machine to the slice corresponding to that bit. This is more in line with the actual needs. Q-learning is one of the classical and effective algorithms in value-based RL methods, which can update the value function in the following iterative way to obtain the optimal policy:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_t + \gamma \max_{a_{t+1}} Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)), \tag{4}$$

where $Q(s_t, a_t)$ is a state–action value function that evaluates the expected reward after action a_t is taken in the state s_t, α describes the rate of Q-value update.

In order to avoid the storage and computational complexity caused by large state and action spaces, deep neural network (DNN) is adopted to approximate the complex non-linear value function, namely $Q(s, a; \delta) \approx Q(s, a)$, where δ is the neural network parameter. This approach is also known as DQL. In addition, two mechanisms are involved to improve the performance of the DQL-based slice admission control scheme. The first is experience replay, where items in the experience replay buffer are sampled in mini-batches to reduce the correlation between training experiences. Another mechanism is the dual DNN structure, where $Q(s, a; \delta)$ and $Q'(s, a; \delta')$ are the output of the evaluation and the target networks respectively, and the network parameters are asynchronously updated by minimizing the mean square error (MSE) loss function as follows:

$$L(\delta) = \mathbb{E}[(r_j + \gamma \max_{a_{j+1}} Q(s_{j+1}, a_{j+1}; \delta') - Q(s_j, a_j; \delta))^2]. \tag{5}$$

Algorithm 1 DRL-based slice admission control

Input: greedy factor ϵ, replay buffer size F, target Q-network update frequency U
Output: trained model, slice admission decision

initialize experience replay buffer E with size F
initialize two Q-networks with parameters δ and δ'
for each episode o **do**
  obtain the DT virtual environment state
  for each step t **do**
    generate a random number $\kappa \in [0, 1]$
    if $\kappa < \epsilon$ **then**
      choose a random action a_t
    else
      choose $a_t = \arg\max_a Q(s_t, a; \delta)$
    end if
    perform attempt in Algorithm 2 based on a_t
    obtain reward r_t and next state s_{t+1}
    store experience item $\{s_t, a_t, r_t, s_{t+1}\}$ in buffer E
    sample mini-batch of transitions $\{s_j, a_j, r_j, s_{j+1}\}$ from buffer E
    if o terminates at step $j+1$ **then**
      $y_j = r_j$
    else
      $y_j = r_j + \gamma \max_{a_{j+1}} Q(s_{j+1}, a_{j+1}; \delta')$
    end if
    perform gradient descent using $(y_j - Q(s_j, a_j; \delta))^2$ as loss
    set $\delta' = \delta$ every U steps
  end for
end for

This mechanism is used to ensure the weak correlation between the evaluation and the target Q-value, and to improve the learning efficiency of the slice admission control scheme.

State, action, and reward are three critical factors in DQL-based algorithms, so we first define them for constructing the DQL-based slice admission control scheme.

- **State**: The state consists of five parts. The first two parts reflect the resource status of the physical core network, including the available amount of processing and transmission resources represented by M_p and M_b. The third part of the state is used to denote the type of slice requests in a processing window, which can be denoted as $N_s = [n_1, n_2, \ldots, n_W], n \in N$. Each n also corresponds to the slice graph G^n, the deployment revenue, and the λ_n of a certain slice. The fourth part represents the admission proportion of each slice request in a processing window, which can be denoted as $H = [h_1, h_2, \ldots, h_W], 0 \le h \le 1$. We define $D = [d_1, d_2, \ldots, d_W], 0 \le d \le 1$ as the decision set for each request in the processing window. In the following, we also give the corresponding mapping for the agent to make a decision d_w on the slice request acceptance condition based on h_w.

$$d_w = \begin{cases} 0 \text{ (reject)}, & h_w < \lambda_{n_w}, \\ h_w \text{ (partially accept)}, & \lambda_{n_w} \le h_w < 1, \\ 1 \text{ (completely accept)}, & h_w \ge 1. \end{cases} \tag{6}$$

  The last part of state is the current slice deployment revenue RV_c. Thus, the state can be expressed as $s = [M_p, M_b, N_s, H, RV_c]$. Specifically, the state dimension dim_s can be represented as $dim_s = 3 + 2W$.
- **Action**: The action indicates the flexible adjustment of the admission proportion of each slice requests in a processing window, $a = [\phi_1, \phi_2, \ldots, \phi_W], \phi \in \{-1, 0, 1\}$. $\phi_w = -1$ and 1 mean that the admission proportion h_w is decreased and increased by a certain amount, which can be defined in advance depending on the circumstances. When $\phi_w = 0$, it means that h_w remains unchanged.
- **Reward**: The target of the DQL algorithm is to maximize cumulative rewards, and we aim to make a series of slice admission decisions with the expectation that these decisions will maximize RV. Therefore, we set the reward to be consistent with our goal as mentioned in Section 3.1, that is, set r as the current deployment revenue RV_c.

Algorithm 2 Slice deployment attempt

Input: action a_t, state s_t
Output: reward r_t, next state s_{t+1}

initialize slice acceptance decision D and available resource condition C from a_t and s_t
for each slice request sr_w in processing window **do**
  obtain slice type n, slice graph G^n, required processing resource ρ and transmission resource β, and acceptance decision $d_w \in D$
  if $d_w > 0$ **then**
    for each VNF v^n in slice **do**
      attempt to deploy v^n on the node with the most abundant processing resource according to d_w
      if deploy v^n successfully **then**
        update the available resource condition C
      else
        restore condition C to its initial state
        calculate and return r_t and s_{t+1}
      end if
    end for
    for each virtual link l^n in slice **do**
      obtain the shortest path sp between the source src and destination dst physical node of l^n
      attempt to deploy l^n on sp according to d_w
      if deploy l^n successfully **then**
        update the available resource condition C
      else
        restore condition C to its initial state
        calculate and return r_t and s_{t+1}
      end if
    end for
  end if
end for
calculate and return r_t and s_{t+1}

Our proposed DRL-based FSAC scheme explores and learns in the DT virtual environment in the form shown in Fig. 2, and the final policy obtained is updated to the physical network through the interaction between the DT plane and the physical entity plane as shown in Fig. 1, avoiding the damage caused by direct exploration on the physical network. Specifically, the proposed slice admission control scheme is shown in Algorithm 1. After the action is obtained in line 12, Algorithm 2 is executed to make slice deployment attempt based on the action. Here, Algorithm 2 provides a processing resource-oriented greedy algorithm, that is, first select the node with the most abundant processing resources for each VNF in the slice to be deployed, and then choose the shortest path for each virtual link. Algorithm 2 also generates new states and rewards based on the results of the slice deployment attempt. Actually, there are a variety of options for slice deployment attempts that can be replaced as needed. In this paper, in addition to the VNF level fine-grained deployment in Algorithm 2, we also apply the concept of resource pool, treating the processing resource and transmission resource in the network as resource pools, and carry out the coarse-grained slice deployment attempt at the resource pool level. The results are presented in the next section.

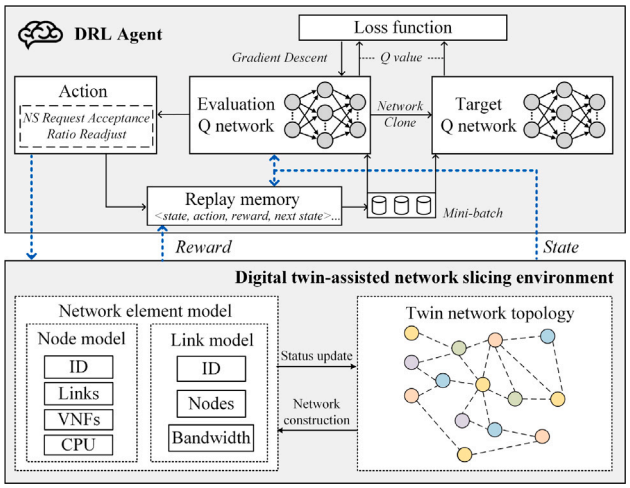


Fig. 2. Illustration of the DRL-based slice admission control scheme for 5G core network.

3.4. Discussions

If trained directly in a real environment, the exploratory actions of RL can have serious adverse effects on physical systems. To this end, we introduce DT technology and build slice admission control scheme based on it. However, as a developing emerging technology, it also has some limitations. Firstly, creating a DT environment is an expensive, resource-driven process. Secondly, the construction of the DT environment relies on open, real-time and high-quality data. Reliable data sources and standard data formats are all places that need attention. The sensitive, real-world system data involved also presents privacy and security risks. Thirdly, DT technology lacks a widely agreed industry standard and requires the concerted efforts of researchers. Therefore, the advantages and difficulties of applying DT technology need to be comprehensively considered. Since the establishment of 5G DT platform is a complex engineering problem, most of the existing excellent DT-related works propose advanced mechanisms based on the architecture, concepts and advantages of DT, and implement them in the way of programming simulation, including some work related to network slicing [25,32] and resource allocation [22,33]. Similarly, the main contribution of this paper is to take advantage of the DT framework and construct a flexible slice admission control scheme, focusing on the general features of the DT technology rather than the specific implementation details. According to the development status of DT technology and our focus, referring to excellent existing literature, the general programming simulation method is also adopted in this paper.

In addition, there exist potential solutions such as safe RL and offline RL that can avoid unsafe actions on the system [34,35]. However, they are not entirely suitable for the scenario we consider. Safe RL is able to make the agent consider the boundary of behavior instead of making arbitrary exploration actions and ensure reasonable system performance. Note that it can only avoid risks with constraints or manual intervention, and cannot directly obtain the optimal solution. There are still cases of system performance degradation. What we want to do is not only avoid compromising the system, but also ensure that the system performance is as optimal as possible. Offline RL means that the system learns from the collected state transition trajectory data without interacting with the environment. Compared to the online RL that we used, there are also some limitations. The first is the problem of data quality. The success of offline RL depends on powerful and accurate trajectory datasets. It is difficult to collect massive real and multi-angle trajectories, and the trajectory generated entirely by expert data or random data may not be effective. The second is the lack of exploration. In RL, exploration is to collect more information and try something impossible. The offline RL completely relies on static datasets, and may not necessarily learn the best policy.

4. Performance evaluation

4.1. Experimental settings

Three representative slice types in 5G are considered in our experiments, namely eMBB, URLLC, and mMTC. The structure of the three types of slices refers to the settings in [6] and is shown in the left part of Fig. 1. Each slice contains the basic 5G core network functions including AMF, session management function (SMF) and user plane function (UPF). The backups of AMF, SMF, and UPF are considered for URLLC slices due to their high-reliability requirements. There are more AMFs in mMTC slice due to the need to handle access of a large number of devices. Specifically, the eMBB slice contains a total of four VNF, namely two AMFs, one SMF and one UPF; the URLLC slice contains one AMF, one SMF, and one UPF, as well as four corresponding VNF backups to improve reliability; the mMTC slice contains three AMFs, one SMF, and one UPF. The requirement for processing resources is set to 10 units for each VNF, except for the backup VNF, which is set to 5 units. In terms of transmission resource consumption, the requirement of each virtual link in eMBB, URLLC, and mMTC slices is set to 3, 2, and 1 units. The slice scaling factor is set to 0.7, 1, and 0.5, and the per-slice deployment profit is set to 2, 1.5, and 1 for three types of slices.

For slice requests, its arrival follows Poisson distribution with the arrival rate of 0.1, and the slice type in the incoming request is random. For the topology of the physical network, we take six topologies of different sizes from Topology Zoo [36] for evaluation, namely Arpanet (4 nodes and 4 edges), Getnet (7 nodes and 8 edges), Abilene (11 nodes and 14 edges), Easynet (19 nodes and 29 edges), Arn (30 nodes and 29 links), and Cesnet (52 nodes and 63 links). Since slice admission control occurs when resources are scarce, except for the topology size evaluation experiment, we use the Getnet topology by default and do not use particularly large-scale topologies. Processing capacity refers to the available processing resource of physical nodes, and transmission capacity refers to the available transmission resource of physical links, both expressed in resource units. By default, we set the processing and transmission capacities as 25 units per node and 20 units per link, the size of the request processing window as 5, and the additional factor as 1. These default values only change if the horizontal coordinates of the experimental figures are changed. We use Python 3.9 and PyTorch to build the slice admission control scheme based on the DQL algorithm. Specifically, the evaluation and the target networks in DQL are two DNNs with the same structure, which have two hidden layers with 50 neurons each. The mini-batch and replay memory sizes are set as 256 and 1000. The target network update frequency is 200. The learning rate and the discount coefficient are set to 0.001 and 0.8, while the ϵ-greedy factor is initially 1 and gradually decreases and stabilizes to 0.1. The detailed DQL settings are summarized in Table 1.

We use the three metrics mentioned in Section 2.4, i.e., the slice acceptance ratio, the slice deployment revenue, and the blocking ratio of unacceptable decisions, to evaluate the performance of the proposed scheme. The comparison schemes include: (1) the proposed DRL-based FSAC scheme with resource pool level coarse-grained slice deployment attempt (FSAC (coarse)), which treats the processing and transmission network resources as resource pools; (2) the proposed DRL-based FSAC scheme with fine-grained VNF level slice deployment attempt (FSAC (fine)), where the detailed attempt is depicted in Algorithm 2; (3) the DRL-based complete admission control (RLCAC) scheme inspired by literature [6] that focus on the complete acceptance or rejection of the slice requests; (4) the first-come-first-served (FCFS) scheme that processes slice requests in the order of their arrival. Note that the experimental results shown in the figures (except for Fig. 8 that focuses on the convergence of the algorithm) are the average of 10 runs of these schemes, each with different slice request arrivals. For DRL-based FSAC and RLCAC schemes, each run contains 1000 episodes.

Table 1
Summary of DQL settings.

Parameter	Value
Number of hidden layers in DNN	2
Number of neurons in hidden layer	50
Mini-batch size	256
Replay memory size	1000
Target network update frequency	200
Learning rate	0.001
ϵ-greedy factor	Gradually decrease and stabilize to 0.1
Discount coefficient	0.8

4.2. Experimental results

In Figs. 3 and 4, we explore the impact of different resource capabilities on the performance of our proposed DRL-based FSAC scheme. Here, the left and right ordinates represent the first two performance metrics we define in Section 2.4. The left ordinate is the bar chart coordinate, representing the slice deployment revenue, while the right ordinate is the line chart coordinate, representing the slice acceptance ratio. In both figures, we can see that slice deployment revenue and slice acceptance ratio increase as processing and transmission capacities increase. Since the slice request processing window is fixed, when the resource capacities increase to a certain extent, the two evaluation indicators tend to be stable, and the differences between various schemes gradually become smaller. It can be seen that the proposed scheme has significant advantages over the other comparison schemes, since it can selectively scale down the requested resource of the slice request to perform partial admission and make full use of network resources. As we can also see, the proposed scheme performs better with resource pool level coarse-grained slice deployment attempts, because slice deployment is less constrained when the resources in the network are treated as resource pools. When a VNF level fine-grained slice deployment attempt is made, physical nodes may have a small amount of remaining resource that is difficult to use. Moreover, through the comparison of RLCAC and FCFS, it can be seen that under similar slice acceptance ratio, RLCAC can get more revenue, which shows the decision-making advantage of DRL-based schemes.

In Fig. 5, we investigate the effect of different slice request processing window sizes on the performance of the proposed slice admission control scheme for 5G core network slicing, where the total amount of resources used for evaluation is fixed. It can be seen that as the window size increases, the revenue gradually increases and becomes stable, because the total amount of resources is fixed, and the maximum number of slices that can be deployed is also limited. For the slice acceptance ratio, it decreases as the slice request processing window size increases, since it is calculated with the window size as the denominator, and its numerator has an upper limit, so it will gradually become smaller. In Fig. 6, we explore the effects of different additional gain factors on revenue. It can be seen that with the increase of additional gain factor, the gap between our proposed scheme and the other two comparison schemes gradually becomes larger. This is because the advantage of our scheme is that more slices can be accepted and deployed, and therefore more additional gain is generated.

In Fig. 7, we analyze the impact of different physical network topologies on slice deployment revenue. Here, we adopt six different sizes of topologies introduced in Section 4.1 to see the scalability of the proposed scheme. In Fig. 7(a), we fix the processing capacity of a single physical node to 25 units to evaluate the performance of the comparison schemes. We can see that with the expansion of the topology scale, the revenue is getting higher. This is because the number of physical nodes and edges rises, leading to an increase in the overall resource of the physical network, and thus more slice deployments can be accommodated. We can also see that when the topology size grows to a certain extent, the revenue does not increase,

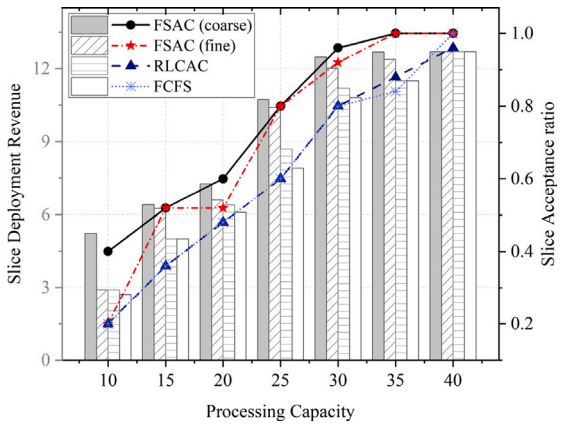


Fig. 3. Performance of the proposed slice admission control scheme under different processing capacities.

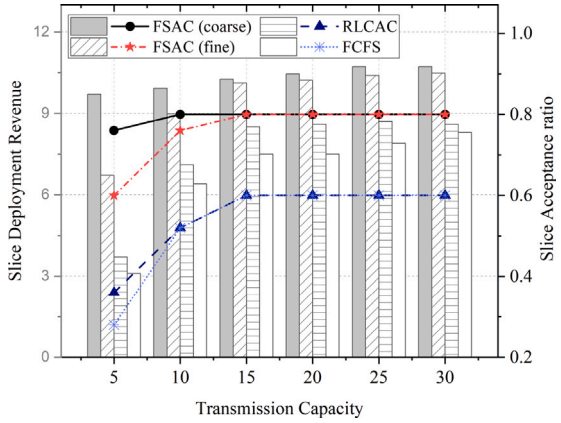


Fig. 4. Performance of the proposed slice admission control scheme under different transmission capacities.

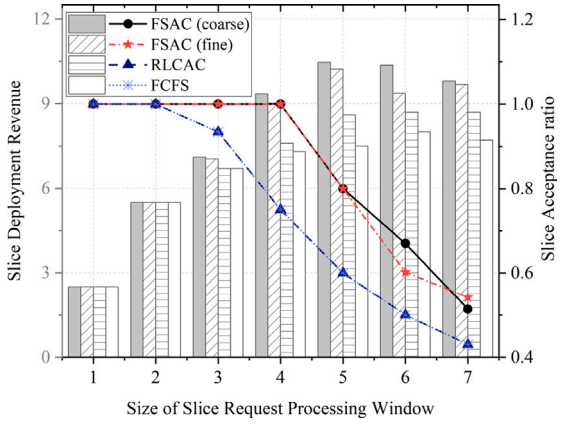


Fig. 5. Performance of the proposed slice admission control scheme under different sizes of slice request processing window.

and the results of several comparison schemes tend to coincide. This is because at this time, network resources are sufficient to fully meet the needs of slice requests, and a simple strategy can achieve good results, leading to the similar performance of all comparison schemes. Different from the processing capacity setting in Fig. 7(a), in Fig. 7(b), the total amount of processing resources in the network is fixed to 200 units and equally allocated to each network node. In this case, as the topology size increases, the resources allocated to each node

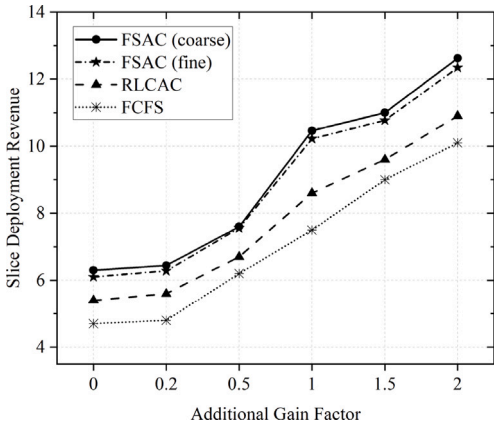
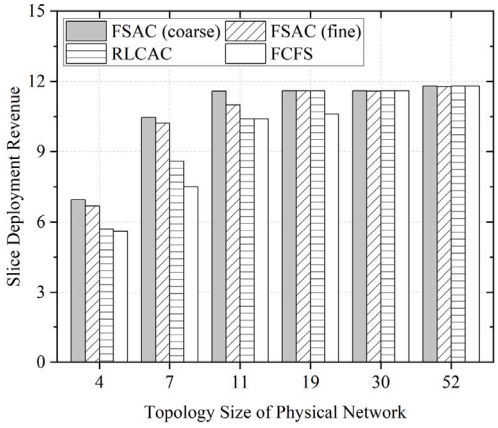


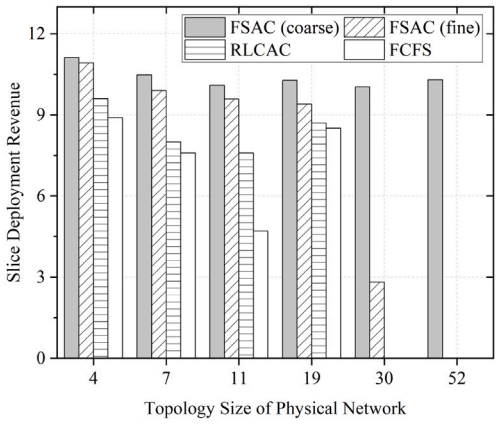
Fig. 6. Slice deployment revenue of the proposed slice admission control scheme under different additional gain factors.

become more fragmented, and the revenue shows a decreasing trend. Especially, in the 52-node topology, each node can only be allocated about 3.85 (200/52) units of resources, making it difficult to implement VNF deployment, so most of the schemes become ineffective. It is worth noting that the performance of FSAC (coarse) changes little in Fig. 7(b), since it treats the overall processing resource as a resource pool to perform allocation, while the amount of overall processing resource does not change in this set of experiments. To some extent, the scheme performance is also related to the matching degree between the nodes' resource and the resource requirement of the slice request. For example, the resource allocated to each node is about 10.53 (200/19) units in the 19-node topology, while in the slice request setup, lots of the VNFs' processing resource requirements are 10 units, which matches the nodes' resource quota, so there are fewer fragmented, hard-to-utilize resources in this topology, and the overall revenue is relatively high. Under different topology scales, it can be seen that our proposed scheme can achieve good results.

In Fig. 8, we evaluate the convergence performance of the proposed DRL-based FSAC scheme. It can be seen that the proposed scheme converges quickly with both coarse- and fine-grained slice deployment attempts. In addition, we change the network resource and the slice request status at 1500 and 3000 episodes, and we can see that the proposed scheme has good flexibility and adaptability to environmental changes, and is able to quickly converge under new environments. For DT application evaluation, in Fig. 9, we investigate the blocking ratio of unacceptable decisions (see Eq. (3)) of the proposed framework under different processing capacities. Here, we adopt the first 500 episodes of each training for evaluation ($EP = 500$), and the unacceptability of a decision is judged by whether the revenue obtained by the decision reaches 70% of the optimal revenue. We can see that in both FSAC (coarse) and FSAC (fine) schemes, the blocking ratios show a trend of decreasing first and then rising slightly. This is because when the processing resource is scarce, it is difficult for DRL agent to get high rewards when exploring the environment, and lots of "bad" decisions will be made before the optimal policy is obtained. When the resource is sufficient, the DRL agent has more choices and possibilities to improve revenue, and the ratio of making unacceptable decisions before getting the optimal strategy is also slightly increased. Moreover, we can see that under different resource capacities, the ratios of unacceptable decisions are all above 25%. This is where the DT environment comes into play. The DT-assisted slice admission control framework is proposed precisely to prevent the impact of these unacceptable decisions on the 5G physical core network.



(a) The processing capacity of a single node is fixed.



(b) The total processing capacity of the network is fixed.

Fig. 7. Slice deployment revenue of the proposed slice admission control scheme in various topologies.

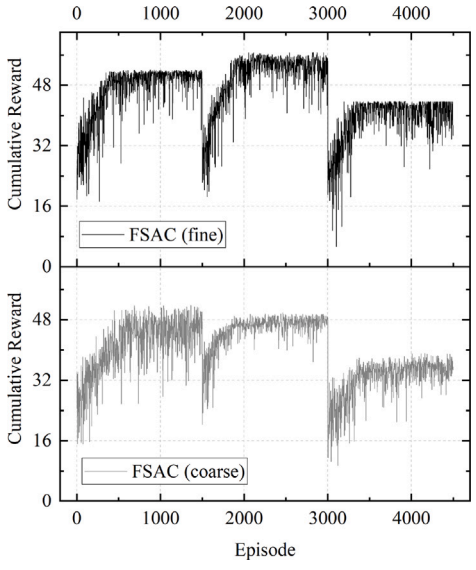


Fig. 8. Convergence and the adaptivity performance of the proposed scheme.

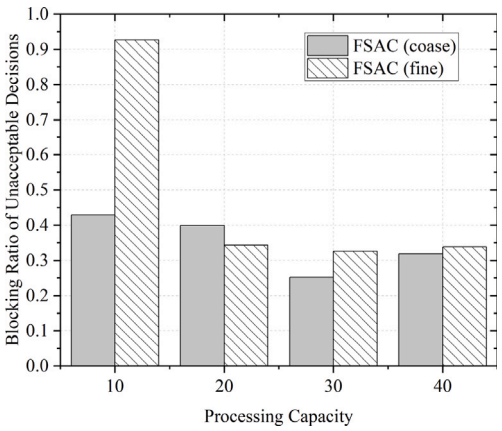


Fig. 9. Blocking ratio of unacceptable decisions under various processing capacities.

5. Conclusion

In this paper, we have studied the slice admission control for 5G core network slicing. Firstly, we designed slice scaling factor to limit the resource adjustment range, and proposed a DRL-based partial admission control scheme to flexibly adjust the slice requested resources and improve the revenue of slice deployment. Secondly, we constructed a DT-assisted slice admission control framework for 5G core network, in order to create a virtual environment for DRL agent to explore and prevent the negative impact on real physical network. Finally, the experimental results showed that our proposed scheme had significant advantages in improving the revenue and slice acceptance ratio. In future work, we will continue to study the intelligent slice admission control schemes in more complex conditions, and build DT platform based on real networks and devices, hoping to make more contributions to the development and wide application of network slicing.

CRediT authorship contribution statement

Jiadai Wang: Conceptualization, Methodology, Software. **Jingyi Li:** Validation, Writing – review & editing. **Jiajia Liu:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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